

# ALLISON DZUBAK

✉ allisondzubak@gmail.com  
🌐 allisondzubak.com  
🔗 allison-dzubak

in allisondzubak  
📞 515-710-8699  
📍 Portland, ME

## Making order out of chaos

Versatile data consultant | Ph.D. in computational chemistry with a track record of turning complex, unstructured problems into efficient, data-driven solutions. Combines technical expertise in algorithm design and automation with experience leading projects, mentoring teams, and translating insights across technical and non-technical audiences. Driven by curiosity and impact, with a collaborative and adaptable approach well-suited to dynamic, mission-driven work.

## SKILLS

➤ **Data Analysis & Coding**  
Python, SQL, Command line, Git, API tools & data integration, ETL, Jupyter, Mathematica, Excel

📊 **Data Visualization & Dashboarding**  
Tableau, Streamlit, plotly, seaborn, matplotlib

☁ **Cloud Computing**  
AWS, Google Colab

⚙ **Machine Learning & AI**  
Dimensionality Reduction, Feature Engineering, Clustering, Regression, Ensemble Methods, Deep Transfer Learning

🌐 **Web Design**  
HTML, CSS, Bootstrap

💻 **High-Performance Computing**  
Large-scale quantum chemical simulation job management

⚙ **Algorithm Development**  
Monte Carlo Methods, Coarse Graining, Optimization Techniques

✍ **Technical Writing**  
Grant proposals, Software documentation, Research highlight slide decks, Academic journal publications

👥 **Collaboration & Communication**  
Microsoft 365, LaTeX, Digital interactive whiteboards, Video creation, Slack, Trello

👤 **Training & Mentoring**  
Universal design for learning, scaffolding, mental health first aid

## EXPERIENCE

### Founder and Data Consultant

#### Katahdin Logic LLC

📅 July 2025 - Present

📍 Portland, ME (Remote)

- Provide backend development, API integration, data analysis, AI (implementation, performance evaluation, optimization), and web-friendly technical content generation services to Civic Minded LLC
- Provide pro bono data cleaning and migration support for Maine Needs local nonprofit

### Assistant Professor of Computational Chemistry

#### Bowdoin College

📅 July 2019 - Feb 2023

📍 Brunswick, ME

- Balanced competing research, teaching, and service demands resulting in a successful reappointment towards tenure
- Developed quantum chemical models, resulting in 14 peer-reviewed publications to date and significant advancements in computational methodologies
- Led a research lab with a \$40K annual budget, guiding 4 undergraduate students in advanced computational projects
- Communicated technical findings by generating visualizations of data, having presented at 20+ conferences and workshops in 5 countries
- Assessed the feasibility and merit of proposed research, resulting in 1 faculty candidate hire and approximately \$270K/year of faculty funding
- Managed and facilitated inclusive learning environments, teaching approximately 150 students and enhancing classroom experiences through smart technology

### Postdoctoral Research Fellow - Materials Science and Technology

#### Oak Ridge National Lab

📅 Feb 2016 - Feb 2019

📍 Oak Ridge, TN

- Secured 1.7 million core hours on Titan supercomputer and 100K core hours on EOS for proposed materials theory research, driving new project exploration
- Executed quantum Monte Carlo simulations on 10 transition metals, quantified key errors, prioritized critical approximations, published in J. Chem. Phys. and supported grant renewal

### Graduate Research Assistant - Department of Chemistry

#### University of Minnesota

📅 2009 - 2015

📍 Minneapolis, MN

- Presented collaborative research to DOE program managers 5 times over 3 years contributing to the \$30 million renewal of the DOE carbon capture center
- Defended Ph.D. research on metal-organic framework computational modeling, receiving the American Chemical Society Chemical Computing Group research excellence award and the department annual award for doctoral thesis excellence

## EDUCATION & CERTIFICATIONS

- Applied Data Science Program: Leveraging AI for Effective Decision-Making [Massachusetts Institute of Technology](#) (2023 - 2024)
- Ph.D. in Computational Chemistry [University of Minnesota](#) (2009 - 2015)
- B.A. in Chemistry [University of Minnesota](#) (2005 - 2009)